

Qianyue Wang - 400385783
Bowen Wang - 400325152
Mian Chen - 400385299
Tianyao Zhang - 400375124
Ethan Suitor - 400411786
Owen Marshall - 400470427
Yicheng Liu - 400359801

Group 5
GhostDrive (Twin)

Verification and Validation for Rev 1

Project Title: *Human-in-the-Loop Cognitive Digital Twin of an Autonomous Scale Vehicle*

Team: Group 5

Supervisor: Dr. Istvan David

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Change log

Date	Version	What is changed	Notes
17/3/2026	1.0	Labels of all the tests are unique now	All test labels should be unique
18/3/2026	1.1	added tests set-up	Add a description of set-up for all the groups of tests.
19/3/2026	1.2	Test description	Brief description of test
20/3/2026	1.3	Removed redundant tests	Some tests overlapped, instead a reference to the overlapped test cases were made
20/3/2026	1.4	Refined tests set-up	

3 Introduction

3.1 Purpose

The purpose of the document is to define the verification and validation (V&V) approach for the digital twin human follower robot system. This document establishes how system requirements are verified and validated through structured test cases and evaluation criteria, ensuring that the system behaves as intended and meets its functional and performance objectives.

3.2 Scope

This document includes a detailed breakdown of our system requirements. It also contains how we plan to test these (our testcases) and breaks them down into input, output and results. Finally, the document includes a description of how our requirements have changed and been updated to align with our verification and validation scheme.

3.3 Background

The digital twin system is being developed to serve as a human follower companion. Its primary purpose is to detect, track and follow a designated human by means of sensing and interpreting the surroundings.

A digital twin architecture is selected in this project to decouple sensing, computation, and control between the physical and virtual systems. This design choice significantly reduces the computational burden on the embedded platform, which typically has limited processing power and energy constraints. By offloading intensive tasks such as object detection and trajectory planning to an external computing unit, the system achieves improved real-time performance, flexibility, and scalability. Additionally, this architecture enables easier debugging, monitoring, and future upgrades without modifying the physical hardware.

Alternative approaches, such as fully onboard processing, were considered; however, these were not selected due to limitations in processing capability, increased power consumption, and reduced ability to iterate rapidly during development. The digital twin approach therefore provides a more efficient and modular solution for prototyping and system validation.

For perception, a YOLO-based object detection model is adopted due to its balance between detection accuracy and real-time inference speed. In comparison to traditional feature-based methods or heavier deep learning models, YOLO offers faster processing with acceptable accuracy, making it suitable for dynamic environments where low latency is critical. This ensures that the system can reliably detect and track a human target while maintaining stable control performance.

Furthermore, the use of a camera-based perception system is chosen over alternative sensing methods such as LiDAR due to its lower cost, lighter weight, and ease of integration with existing vision-based algorithms. While LiDAR provides higher depth accuracy, it is not necessary for this application where relative positioning and visual tracking are sufficient.

The digital twin is a system that runs outside of the vehicle (e.g. on a laptop or in the cloud). The physical twin collects and communicates information about its surroundings to the digital twin, which in turn performs computationally expensive tasks to direct the physical twin in its behavior. The physical twin is the physical human follower vehicle.

4 Validation and Updated Requirements

4.1 Requirement Overhaul

It was decided the original requirements no longer reflect the current scope of the project. This document presents a revised list of requirements which do reflect the current objectives of the system. These were created such that if they were all met, the system would operate as intended. Specifically, the scope of the digital twin system was narrowed to have more simplistic functionality.

The original requirements were ambitious and made specific unrealistic guarantees on aspects such as timing between events. Additionally, they contained ambiguous statements which lack the ability to be verified and generally seemed unfeasible to accomplish with the remaining time.

When revising the requirements, several members iterated through the previous set and determined which ones were no longer reflective, and to what degree. Some requirements were changed, some were removed completely and some additions were made. It is possible there are requirements that have still been overlooked, but it is believed the revised set much better describes the aspects necessary to make the system operate as intended.

4.2 Modified Requirements

Table 1: Table of adjusted requirements.

Old ID(s)	New ID(s)	Nature of Adjustment
DSR1, HW-1 DSR2, PT-2	REQ-SB-11	The 4 old requirements are redundant and too specific for obstacle avoidance which limit our design constraints.
DSR3, HW-5	REQ-SB-05	Simplification of the 2, set a hard speed limit.
DSR4, PT-8, HW-6	REQ-GSB-02	Merged hardware/software stop response into a single requirement.
DSR5	REQ-SB-01	"Unverified regions" are now handled with the scope change. Vehicle will operate only in an indoor environment.
DSR7	REQ-SB-08	"Reliable identification" is now specific to "ignoring others in FOV".
DSR8, PT-5	REQ-ST-05	"Target Lost" is now a specific FSM transition rule to be performed within the state transition process.
DSR9	REQ-ST-09	Old requirement description was convoluted, also did not specify state(s) when requirement was applicable. New requirement specifies what state(s) requirement applies to and clearly outlines the required behaviour.

DSR11	REQ-ST-05	Old requirement was ambiguous in the description of target being lost. Additionally, it did not specify state transition behaviour. New requirement uses clearer language and specifies required state behaviour.
DSR13	REQ-GSB-02	Restructured requirement to focus on digital twin side of communication providing an emergency stop. Old requirement was focusing on halting vehicle if communication is lost, but this is not needed as is no communication is available, no commands can be given anyway, so the physical twin would halt already.
PT-1	REQ-SM-02	Relaxed requirement. Previous requirement required a specific amount of time that may not be achievable in all situations. New requirement specifies and average to accommodate for varying response times under different operation modes.
PT-3, PT-4	REQ-SB-07	Response to gestures/position is now defined by specific SB triggers.
HW-4	REQ-SB-05	Definition simplified into Max Velocity/Steering limits.
SW-3	REQ-SB-09	Timing constraint is unrealistic both in timing and measurability. Changed to measure of confidence.
SW-6	REQ-SM-01	Simplified definition.

4.3 Newly added Requirements

Table 2: Table of added requirements.

New ID	Functional Addition
REQ-SB-02	Command to halt current movement while in Idle state.
REQ-SB-03	User-defined following distance (0.5m to 2.0m).
REQ-SB-04	Ignoring gestures unless in specific active states.
REQ-SB-07	The specific gesture library (Stop, Resume, Far, Close).
REQ-SB-10	Maintain a minimum clearance of 0.2m from obstacles
REQ-ST-01	System power-up default state logic.
REQ-ST-02/03	Strict requirements for exiting the Initial state.

REQ-ST-04	Entry condition for Follow state.
REQ-ST-06/07/08	Entry/Exit and safety override rules for Manual Mode.
REQ-ST-10	Context-aware fallback (returning to Idle vs Initial).
REQ-GSB-03	Mode switching without system restart.
REQ-SM-02	User can visualize system state through display mechanisms via the Digital Twin interface.

4.4 Removed Requirements

Table 3: Removed requirements

Old ID	Reason for Deletion
DSR6	Automation of speed reduction based on frequency is too complex to implement reliably in ROS2.
DSR10	Battery shutdown is now a safety assumption rather than a core functional requirement.
DSR12	Bad requirements, this is not something handleable by the control software.
DSR14	Manual recovery requirements are now covered by REQ-ST-02.
PT-6	Sensor timeout specifically at 0.2s is merged into the general 100ms mirror/200ms stop logic. May still be subject to change, need to verify if these timing requirements are realistic.
PT-7	Explicit logging frequency (5Hz/25Hz) is removed, irrelevant requirement.
HW-2/3/7/8/9	The car is pre-operational, it is not something that we develop the hardware for. We will assume that the hardware is in good working order.
SW-2	1ms system clock resolution is a property of ROS, not a requirement.
SW-4/5/7	QoS config, replay facilities, and test harnesses are infrastructure tools, not system behaviors (too specific and irrelevant).

Table 4: Updated list of requirements.

Requirement Type	Requirement ID	Description
State Behaviour	REQ-SB-01	The system shall not command the PT to move while in the initial state.
	REQ-SB-02	The system shall command the PT to halt current movement while in the idle state.
	REQ-SB-03	The system shall command the PT to follow the target user at a setpoint distance defined by the user (0.5m to 2.0m) while in the Follow state.
	REQ-SB-04	The system shall not respond to motion controlling gestures unless it is inside the Idle or Follow (in future could be more) states.
	REQ-SB-05	The system shall not command the PT to move at a linear or angular velocity above 3 m/s and 1.4 rad/s
	REQ-SB-06	The system shall command the PT to change its linear and angular velocity in accordance with user input while inside the Manual Mode state.
	REQ-SB-07	The system shall support distinct gesture commands to trigger specific behaviors while inside the follow state: Stop, Resume/Proceed, Increase Distance (Far), and Decrease Distance (Close).
	REQ-SB-08	The system shall accept gesture commands only from the tracked user, ignoring gestures from other persons in the field of view.
	REQ-SB-09	The system shall process only the most confidently detected gesture
	REQ-SB-10	The system shall modify its path to maintain a minimum clearance of 0.2m from obstacles
State Transitions	REQ-ST-01	The system shall start in the initial state upon initial powerup or restart.
	REQ-ST-02	The system shall not transition out of the initial state unless a user signals to enter manual mode or displays the tracking gesture to the PT.
	REQ-ST-03	The system shall not transition from the initial state to the idle state unless a user is in sight of and displays the tracking gesture to the PT.
	REQ-ST-04	The system shall not transition to the follow state unless the system is in the idle state and the tracked user provides the follow gesture to the PT.
	REQ-ST-05	The system shall not transition from the follow state to the idle state unless the tracked user is lost or the tracked user provides the stop follow gesture.
	REQ-ST-06	The system shall transition to manual mode from any state given the user provides the manual mode input signal.
	REQ-ST-07	The system shall not transition to manual mode unless the user provides the manual mode input signal.
	REQ-ST-08	The system shall not exit manual mode unless the user provides the manual mode exit signal.
	REQ-ST-09	Provided the exit condition is met, the system shall transition from manual mode to the initial state if there is no current tracked person.

	REQ-ST-10	Provided the exit condition is met, the system shall transition from manual mode to the idle state if there is a current tracked person.
General System Behaviour	REQ-GSB-01	The system shall provide a Manual Mode where the user can directly control the PT through DT
	REQ-GSB-02	The system shall allow the user to interrupt autonomous operation at any time via a hardware or software command, transitioning the vehicle to a STOP (initial) state within 200ms .
	REQ-GSB-03	The system shall enable the user to switch between "Autonomous" and "Manual" modes at any time without restarting the system.
System Monitor	REQ-SM-01	The DT shall provide a visual interface to monitor the real-time state variables.
	REQ-SM-02	The DT shall mirror the states and the monitored variables of the PT with a maximum average latency of 500ms

The requirements that are highlighted in red are the ones that are likely to be changed as we continue the development and the testing process if the system reaches its software or hardware bottleneck.

5 Verification

5.1 Test Plan and Factors

The testing policy for verification of the system requirements consisted of iterating through the list of requirements and creating an exhaustive set of test cases to verify correct behaviour. This involved determining inputs and outputs which could affect the requirement as well as considering edge cases. For some requirements the implementation has not been completed and therefore the test was marked incomplete.

Factors such as lighting condition, different users, proximity of PT vs DT and repeatability were considered. To be specific, the testing environment was an indoor room with dimmable lights, and different people played the role of target person and distractor. The proximity of PT vs DT did not exceed 20m, so further testing will be needed to verify longer range capabilities. Lastly, tests were performed multiple times under each condition to verify integrity of the results.

5.1.1 Gesture ID and Commands

Table 5: Table of Gesture ID and Commands

Gesture ID	Gesture	Command description
Gesture 1	Thumbs down	Stop the PT from following the target user, return to idle state
Gesture 2	Palm	Command the PT to follow the target user, move from idle to follow state
Gesture 3	TBD	Set the following distance closer during the following state
Gesture 4	TBD	Set the following distance further during the following state
Gesture 5	Heart	Tag the target user

5.1.2 Default Controlled Variable Values

Table 6: Table of controlled variables.

Controlled Variable	Default Value	Units
C_turn_cmd	0.0	rad/s
C_forward_cmd	0.0	m/s
C_linear_velocity_x	0.0	m/s
C_angular_velocity_y	0.0	rad/s
C_mm_toggle	0.0	NA
C_mm_forward	0.0	m/s
C_mm_turn	0.0	rad/s
C_emergency_stop	0	N/A

5.1.3 Default Monitored Variable Values

Table 7: Table of monitored variables.

Monitored Variable	Default Value	Units
M_p_class	0	NA
M_g_class	0	NA
M_g_confidence	0.0	%
M_scan	Empty laserscan message	NA

5.1.4 Default User position

Refer to figure 1 in the appendix section:

Table 8: List of Possible Standing Position

P2	The center of the camera view
P1	The left edge of the camera view
P3	The right edge of the camera view

The default testing position where the user will be standing will always be at P2 unless specifically mentioned in the input section



5.1.5 Default gesture position

Refer to Figure 2 in the appendix section:

Table 9: List of Possible Gesture Positions

P1	The center within the YOLO bounding box
P2	The left edge of the YOLO bounding box
P3	The right edge of the YOLO bounding box

5.1.6 Test Coverage Criteria

Testing coverage was determined based on having a test case for each requirement, and for each of those test cases, the test be conducted several times under different conditions as mentioned above.

5.2 Test Set

Testing Classification

The testing process is divided into three levels:

1. Unit Testing

Tests are performed on individual modules in isolation, typically using software-in-the-loop (SIL) on a laptop. Examples include testing the object detection module (YOLO), tracking algorithms, and control components independently. Performance is evaluated using metrics such as detection accuracy (precision/recall), processing latency (ms per frame), and control error (e.g., steady-state error and overshoot).

2. Integration Testing

Tests evaluate interactions between subsystems such as perception, control, and communication. These tests involve partial hardware integration, including the Digital Twin (DT) and Physical Twin (PT) using ROS2. Examples include validating the perception-to-control pipeline, ensuring correct communication between DT and PT, and verifying sensor data synchronization through ROS2 message passing. Performance is assessed based on end-to-end latency, communication delay, packet loss, and system stability during continuous operation.

3. System-Level Testing

Tests are conducted on the complete system, including the physical robot, sensors, and real-world operating environment. Examples include real-time object tracking, human-in-the-loop interaction (e.g., gesture control), and operation under dynamic environmental conditions such as varying lighting or obstacles. Evaluation metrics include tracking accuracy, response time to user inputs, robustness to disturbances (e.g., noise and occlusion), and safety performance such as collision avoidance and fail-safe behavior.

5.2.1 State Behaviour (System-Level Test)

General test setup:

All tests are conducted using a full system setup including the physical robot (PT), camera-based YOLO perception, and ROS2 communication with the Digital Twin (DT) in an indoor environment.

Both physical and digital twin should be connected to the same WLAN.

Physical robot is running broadcasting to transmit video stream,

Digital twin is running yolo detection with python and MATLAB to control the robot.

MATLAB is also used for monitoring which state of the system is in at the moment.

UI software or MATLAB are used to monitor if the human or gesture is detected and trusted.

Tester as user should be inside the camera angle and shown in the captured video.

REQ-SB-01: The system shall not command the PT to move while in the initial state.

Test description: All the possible movement commands shall be input individually while in the initial state to evaluate this requirement.

Test setup: System state shall be set to initial state.

Table 10: REQ-SB-01

TEST CASES	INPUT	INITIAL STATE(S)	EXPECTED OUTPUT STATE	EXPECTED VARIABLE CHANGE(S)	ACTUAL OUTPUT STATE	ACTUAL VARIABLE CHANGE(S)	PASS
SB-01-T01	Manual mode movement controls	Initial	Initial	None	Initial	None	Yes
SB-01-T02	Gesture 1	Initial	Initial	None	Initial	None	Yes
SB-01-T03	Gesture 2	Initial	Initial	None	Initial	None	Yes
SB-01-T04	Gesture 3	Initial	Initial	None	Initial	None	Yes

SB-01-T05	Gesture 4	Initial	Initial	None	Initial	None	Yes
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Pass/Fail Criteria:

A test is considered a Pass if the system remains in the Initial state and no motion command is sent to the PT. Specifically, the velocity must remain zero, and no control signal (command_sent = FALSE) should be transmitted to the PT under any input condition. Any deviation from these conditions is considered a Fail.

REQ-SB-02: The system shall command the PT to halt current movement while in the idle state.

Test Description: All possible transitions to the idle state are tested to evaluate this requirement.

Test setup: system in various states, with change of input with stop gesture(dislike), manual mode toggle off and person lost track in video stream

Table 11: REQ-SB-02

Test Cases	Input	Initial state(s)	Expected output State	Expected Variable Change(s)	Actual Output State	Actual Variable Change(s)	Pass
SB-02-T01	Manual Mode Toggle On to Off	Manual Mode	Idle	C_linear_velocity_x -> 0 C_angular_velocity_z -> 0	Idle	C_linear_velocity_x -> 0 C_angular_velocity_z -> 0	Yes
SB-02-T02	Gesture 1	Follow	Idle	C_linear_velocity_x -> 0 C_angular_velocity_z -> 0	Idle	C_linear_velocity_x -> 0 C_angular_velocity_z -> 0	Yes
SB-02-T03	Person Lost (M_p_class 1 -> 0)	Follow	Idle	C_linear_velocity_x -> 0 C_angular_velocity_z -> 0	Idle	C_linear_velocity_x -> 0 C_angular_velocity_z -> 0	Yes

Pass/Fail Criteria:

A test is considered a Pass if the system transitions to the Idle state and actively commands the PT to stop all motion. This requires both linear and angular velocities to be reduced to zero ($C_linear_velocity_x = 0$, $C_angular_velocity_z = 0$), and a stop command to be issued to the PT. Any residual motion or failure to send a stop command is considered a Fail.

REQ-SB-03: The system shall command the PT to follow the **target user** at a setpoint distance defined by the user (0.5m to 2.0m) while in the Follow state.

Test description: Non-overlapping positions of target user and a distractor in the camera frame tested to evaluate the requirement. Gestures that modify following distance also tested.

Test setup: target user in the frame initially and gestures were made, before and after a distractor entering the frame. Distractor should walk around and in front of camera to try to block the view of the target user from the camera.

Table 12: REQ-SB-03

TEST CASES	INPUT	INITIAL STATE(S)	EXPECTED OUTPUT STATE	EXPECTED VARIABLE CHANGE(S)	ACTUAL OUTPUT STATE	ACTUAL VARIABLE CHANGE(S)	PASS
SB-03-T01	Gesture 3	Follow	Follow	C_distance += 0.25m	Follow	C_distance += 0.25m ± 0.1m	Yes
SB-03-T02	Gesture 4	Follow	Follow	C_distance -= 0.25m	Follow	C_distance -= 0.25m ± 0.1m	Yes
SB-03-T03	Target user Position: P2	Follow	Follow	C_linear_velocity_x = C_forward_cmd	Follow	C_linear_velocity_x = C_forward_cmd	Yes
SB-03-T04	Target user Position: P2 Distractor Position: P3	Follow	Follow	C_linear_velocity_x = C_forward_cmd	Follow	C_linear_velocity_x = C_forward_cmd	Yes
SB-03-T05	Target user Position: P1 Distractor Position: P3	Follow	Follow	C_linear_velocity_x++ C_angular_velocity_z--	Follow	C_linear_velocity_x++ C_angular_velocity_z--	Yes
SB-03-T06	Target user Position: P3 Distractor Position: P1	Follow	Follow	C_linear_velocity_x++ C_angular_velocity_z++	Follow	C_linear_velocity_x++ C_angular_velocity_z++	Yes

Pass/Fail Criteria:

A test is considered a Pass if the system remains in the Follow state and correctly tracks the target user while maintaining the desired setpoint distance within the range of 0.5m to 2.0m. The system must generate appropriate linear and angular velocity commands based on distance and orientation errors. The target must be correctly identified even in the presence of distractors. Any incorrect tracking, unstable motion, or violation of distance constraints is considered a Fail.

REQ-SB-04: The system shall not respond to motion controlling gestures unless it is in the Idle or Follow (in future could be more) states.

Test description: Try all automatic movement gestures in states that are not the idle or follow states to evaluate the requirement.

Test setup: target user in the frame initially and gestures were made.

Table 13: REQ-SB-04

TEST CASES	INPUT	INITIAL STATE(S)	EXPECTED OUTPUT STATE	EXPECTED VARIABLE CHANGE(S)	ACTUAL OUTPUT STATE	ACTUAL VARIABLE CHANGE(S)	PASS
SB-04-T01	Gesture 1	Initial	Initial	None	Initial	None	Yes
SB-04-T02	Gesture 2	Initial	Initial	None	Initial	None	Yes
SB-04-T03	Gesture 3	Initial	Initial	None	Initial	None	Yes
SB-04-T04	Gesture 4	Initial	Initial	None	Initial	None	Yes
SB-04-T05	Gesture 1	Manual Mode	Manual Mode	None	Manual Mode	None	Yes
SB-04-T06	Gesture 2	Manual Mode	Manual Mode	None	Manual Mode	None	Yes
SB-04-T07	Gesture 3	Manual Mode	Manual Mode	None	Manual Mode	None	Yes
SB-04-T08	Gesture 4	Manual Mode	Manual Mode	None	Manual Mode	None	Yes

Pass/Fail Criteria:

A test is considered a Pass if the system remains in its current state (Initial or Manual Mode) and does not execute any motion commands in response to gestures. The system may detect the gesture, but it must not validate it (gesture_valid = FALSE) or send any control commands (command_sent = FALSE). Any unintended motion or state transition is considered a Fail.

REQ-SB-05: The system shall not command the PT to move at a linear or angular velocity above 3 m/s and 1.4 rad/s

Test description: Input largest position movement command from digital twin in linear and angular directions from all possible movement command states to evaluate requirement.

Test setup: system state is set to manual/idle state.

Table 14: REQ-SB-05

TEST CASES	INPUT	INITIAL STATE(S)	EXPECTED OUTPUT STATE	EXPECTED VARIABLE CHANGE(S)	ACTUAL OUTPUT STATE	ACTUAL VARIABLE CHANGE(S)	PASS
SB-05-T01	C_mm_forward = full forward on joystick	Manual	Manual	C_linear_velocity_x: 0 -> 0.2	Manual	C_linear_velocity_x: 0 -> 0.2	Yes
SB-05-T02	C_mm_forward = full backward on joystick	Manual	Manual	C_linear_velocity_x: 0 -> -0.2	Manual	C_linear_velocity_x: 0 -> -0.2	Yes
SB-05-T03	C_mm_turn = full left on joystick	Manual	Manual	C_angular_velocity_z: 0 -> 0.3	Manual	C_angular_velocity_z: 0 -> 0.3	Yes
SB-05-T04	C_mm_turn = full right on joystick	Manual	Manual	C_angular_velocity_z: 0 -> -0.3	Manual	C_angular_velocity_z: 0 -> -0.3	Yes
SB-05-T05	Gesture 2	Idle	Follow	C_linear_velocity_x = C_forward_cmd (0.1) C_angular_velocity_z = C_turn_cmd (0.2)	Follow	C_linear_velocity_x = C_forward_cmd (0.1) C_angular_velocity_z = C_turn_cmd (0.2)	Yes

Pass/Fail Criteria:

A test is considered a Pass if the commanded linear and angular velocities never exceed the maximum limits of 3 m/s and 1.4 rad/s, respectively, under any input condition. The system must enforce saturation (clipping) of velocity commands when inputs exceed allowable limits. Any instance where the commanded velocity exceeds these thresholds is considered a Fail.

REQ-SB-06: The system shall command the PT to change its linear and angular velocity in accordance with user input while inside the Manual Mode state.

Test description: Input various levels of position movement command from digital twin in manual mode to evaluate requirement.

Test setup: system state is set to manual state.

Table 15: REQ-SB-06

TEST CASES	INPUT	INITIAL STATE(S)	EXPECTED OUTPUT STATE	EXPECTED VARIABLE CHANGE(S)	ACTUAL OUTPUT STATE	ACTUAL VARIABLE CHANGE(S)	PASS
SB-06-T01	C_mm_forward= no input to joystick	Manual	Manual	C_linear_velocity_x: 0 -> 0	Manual	C_linear_velocity_x: 0 -> 0	Yes
SB-06-T02	C_mm_forward = half forward on joystick	Manual	Manual	C_linear_velocity_x: 0 -> 0.1	Manual	C_linear_velocity_x: 0 -> 0.1	Yes
SB-06-T03	C_mm_forward= half backward on joystick	Manual	Manual	C_linear_velocity_x: 0 -> -0.1	Manual	C_linear_velocity_x: 0 -> -0.1	
SB-06-T04	C_mm_turn = no input to joystick	Manual	Manual	C_angular_velocity_z: 0 -> 0	Manual	C_angular_velocity_z: 0 -> 0	
SB-06-T05	C_mm_turn = half left on joystick	Manual	Manual	C_angular_velocity_z: 0 -> 0.15	Manual	C_angular_velocity_z: 0 -> 0.15	Yes
SB-06-T06	C_mm_turn = half right on joystick	Manual	Manual	C_angular_velocity_z: 0 -> -0.15	Manual	C_angular_velocity_z: 0 -> -0.15	Yes

Pass/Fail Criteria:

A test is considered a Pass if the system remains in Manual Mode and generates linear and angular velocity commands proportional to the user input. Zero input must result in zero velocity, and increasing input magnitude must result in a corresponding increase in velocity. Any mismatch between input and output response is considered a Fail. This test verifies that the system implements a proportional control mapping between user input and commanded velocity in Manual Mode.

REQ-SB-07: The system shall support distinct gesture commands to trigger specific behaviors while inside the idle or follow states: Stop, Resume/Proceed, **Increase Distance (Far)**, and **Decrease Distance (Close)**.

Test description: Input all various gesture commands while in the idle or follow states to evaluate requirement.

Test setup: user is identified and tracked.

Table 16: REQ-SB-07

TEST CASES	INPUT	INITIAL STATE(S)	EXPECTED OUTPUT STATE	EXPECTED VARIABLE CHANGE(S)	ACTUAL OUTPUT STATE	ACTUAL VARIABLE CHANGE(S)	PASS
SB-07-T01	Gesture 2	Idle	Follow	C_linear_velocity_x: 0 -> 0.2	Follow	C_linear_velocity_x: 0 -> 0.2	Yes
SB-07-T02	Gesture 1	Follow	Idle	C_linear_velocity_x: 0.2 -> 0	Idle	C_linear_velocity_x: 0.2 -> 0	Yes
SB-07-T03	Gesture 3	Follow	Follow	C_distance += 0.25m			Incomplete
SB-07-T04	Gesture 4	Follow	Follow	C_distance -= 0.25m			Incomplete

Pass/Fail Criteria:

A test is considered a Pass if each gesture is correctly recognized and triggers its corresponding behavior without ambiguity. The system must map each gesture to a unique action: Stop (halt motion), Resume (enter Follow state), Increase Distance, and Decrease Distance. Any incorrect mapping, missed detection, or unintended behavior is considered a Fail. This test verifies correct gesture-to-action mapping and ensures unambiguous control behavior in gesture-based interaction.

REQ-SB-08: The system shall accept gesture commands only from the tracked user, ignoring gestures from other persons in the field of view.

Test description: Give gesture commands in various positions relative to bounding box of target person to evaluate requirement.

Test setup:

System shall be set to the Idle or Follow state as required by the test case. A target user shall first be identified and tracked by the system so that a valid YOLO person bounding box is established. A second person (non-target user) shall then enter the camera field of view and perform gesture

commands at different positions relative to the tracked user's bounding box, including outside the bounding box, at the left edge, at the right edge, and inside the bounding box.

Table 17: REQ-SB-08

TEST CASES	INPUT	INITIAL STATE(S)	EXPECTED OUTPUT STATE	EXPECTED VARIABLE CHANGE(S)	ACTUAL OUTPUT STATE	ACTUAL VARIABLE CHANGE(S)	PASS
SB-08-T01	Outside bounding box Gesture 2	Idle	Idle	None	Follow	None	Yes
SB-08-T02	Non-target person gesture on the outside left edge of the target person bounding box, Gesture 2	Idle	Idle	None	Idle	None	Yes
SB-08-T03	Non-target person gesture on the outside right edge of the target person bounding box, Gesture 2	Idle	Idle	None	Idle	None	Yes
SB-08-T04	Non-target person gesture inside bounding box, Gesture 2	Idle	Idle	None	Follow	C_linear_velocity_x: 0 -> 0.2	No
SB-08-T05	Non-target person gesture inside bounding box, Gesture 1	Follow	Follow	None	Idle	C_linear_velocity_x: -> 0	No

Pass/Fail Criteria:

A test is considered a Pass if the system only accepts gesture commands from the tracked target user. Gestures from non-target individuals must be detected but ignored (gesture_valid = FALSE, command_sent = FALSE), and no state transition or motion command should occur. Any response

to a non-target gesture is considered a Fail. This test ensures robust user-specific interaction by validating that gesture commands are associated only with the tracked target identity.

REQ-SB-09: The system shall process only the most confidently detected gesture

Test description: Provide multiple gestures inside target person bounding box to evaluate requirement.

Test setup: user is tracked and multiple gestures are made at the same time.

Table 18: REQ-SB-09

TEST CASES	INPUT	INITIAL STATE(S)	EXPECTED OUTPUT STATE	EXPECTED VARIABLE CHANGE(S)	ACTUAL OUTPUT STATE	ACTUAL VARIABLE CHANGE(S)	PASS
SB-09-T01	Gesture 2 (precisely shown) and gesture 1 (not precisely shown) inside the target person bounding box	Idle	Follow	C_linear_velocity_x: 0 -> 0.2	Follow	C_linear_velocity_x: 0 -> 0.2	Yes
SB-09-T02	Gesture 1 (precisely shown) and gesture 2 (not precisely shown) inside the target person bounding box	Follow	Idle	C_linear_velocity_x: -> 0	Idle	C_linear_velocity_x: -> 0	Yes

Pass/Fail Criteria :

A test is considered a Pass if, when multiple gestures are detected simultaneously, the system processes only the gesture with the highest confidence score and ignores all lower-confidence gesture candidates. The resulting state transition and motion command must correspond only to the selected gesture. Any mixing of commands or response to a lower-confidence gesture is considered a Fail. This test verifies deterministic gesture arbitration under ambiguous multi-gesture conditions.

REQ-SB-10: **The system shall modify its path to maintain a minimum clearance of 0.2m from obstacles**

Test description: Move the DT close to obstacles in manual mode to evaluate the requirement. Also move obstacles in front of DT.

Test setup: System shall be placed in the Follow state with a valid tracked user ahead of the PT, or in a moving condition as required by the test case. Static and dynamic obstacles shall be introduced into the robot path, including walls, small objects, and suddenly placed obstacles. Distances shall be measured approximately in the physical environment to confirm when objects are within the 0.2 m clearance region.

Table 19: REQ-SB-10

TEST CASES	INPUT	INITIAL STATE(S)	EXPECTED OUTPUT STATE	EXPECTED VARIABLE CHANGE(S)	ACTUAL OUTPUT STATE	ACTUAL VARIABLE CHANGE(S)	PASS
SB-10-T01	Wall in between of user and PT	Follow	Follow	C_linear_velocity_x: 0.2 -> 0			Incomplete
SB-10-T02	Place a small obstacle directly on the path to the target.	Follow	Follow	C_linear_velocity_x: -- C_angular_velocity_z: ++			Incomplete
SB-10-T03	While robot is moving, slide an object into its path.	Follow	Follow	C_linear_velocity_x: -- C_angular_velocity_z: ++			Incomplete
SB-10-T04	Place robot physically 0.1m from a wall	Follow	Follow	0.2 -> 0			Incomplete

Pass/Fail Criteria:

A test is considered a Pass if the system modifies its motion to avoid obstacles and maintains a minimum clearance of 0.2 m whenever possible. The system must reduce forward velocity, stop, or adjust angular velocity as needed to prevent collision. Any collision, continued approach into unsafe distance, or failure to react to an obstacle is considered a Fail.

5.2.2 State Transition (System-Level test)

REQ-ST-01: The system shall start in the initial state upon initial powerup or restart.

Test description: Power up and restart system to evaluate the requirement

Test setup: For the power-up test, the system shall be completely powered off before startup. For the restart test, the running DT/PT software stack shall be stopped and relaunched, or the system shall be rebooted.

Table 20: REQ-ST-01

TEST CASES	INPUT	INITIAL STATE(S)	EXPECTED OUTPUT STATE	EXPECTED VARIABLE CHANGE(S)	ACTUAL OUTPUT STATE	ACTUAL VARIABLE CHANGE(S)	PASS
ST-01-T01	Restart	-	Initial	None	Initial	None	Yes
ST-01-T02	Power up	-	Initial	None	Initial	None	Yes

Pass/Fail Criteria :

A test is considered a Pass if, upon power-up or restart, the system initializes into the Initial state with all motion commands reset to zero and no control signals sent to the PT. Any incorrect initial state or residual motion commands is considered a Fail. This test verifies deterministic system initialization and ensures a safe startup condition.

REQ-ST-02: The system shall not transition out of the initial state unless a user signals to enter manual mode or displays the tracking gesture to the PT.

Test description: Provide a variety of commands to the system to evaluate the requirement.

Test setup: Different command inputs shall then be applied individually, including manual mode toggle, tracking gesture, other gestures, and joystick/manual movement commands. For cases where “user in frame” is required, the target user shall be clearly visible to the camera.

Table 21: REQ-ST-02

REQ-ST-03: The system shall not transition from the initial state to the idle state unless a user is in sight of and displays the tracking gesture to the PT.

Test Description: Provide various gestures and command inputs to show transition only occurs for correct scenario.

Test setup: general setup.

Table 22: REQ-ST-03

TEST CASES	INPUT	INITIAL STATE(S)	EXPECTED OUTPUT STATE	EXPECTED VARIABLE CHANGE(S)	ACTUAL OUTPUT STATE	ACTUAL VARIABLE CHANGE(S)	PASS
ST-02-T01	Manual mode toggle off to on	Initial	Manual Mode	C_mm_toggle 0 -> 1	Manual Mode	C_mm_toggle 0 -> 1	Yes
ST-02-T02	Gesture 5 and user in frame	Initial	Idle	M_p_class 0 -> 1	Idle	M_p_class 0 -> 1	Yes
ST-02-T03	Gesture 2	Initial	Initial	None	Initial	None	Yes
TEST CASES	Input	Initial state(s)	Expected output State	Expected Variable Change(s)	Actual Output State	Actual Variable Change(s)	Pass
ST-03-T01	Gestures 1,2,3 and 4 and user in frame	Initial	Initial	None	Initial	None	Yes
ST-03-T02	Gesture 5 and user in frame	Initial	Idle	M_p_class 0 -> 1	Idle	M_p_class 0 -> 1	Yes
ST-03-T03	C_forward_cmd = various joystick positions	Initial	Initial	None	Initial	None	Yes

and user in
frame

Pass/Fail Criteria :

A test is considered a Pass if the system transitions from the Initial state to the Idle state only when a valid tracking gesture is detected and a user is present in the field of view. Any other gesture or input must not trigger a transition. Any unintended transition is considered a Fail.

REQ-ST-04: The system shall not transition to the follow state unless the system is in the idle state and the tracked user provides the follow gesture to the PT.

Test description: Provide various commands while in the idle state to evaluate the requirement.

Test setup: System state shall be set to the Idle state with a target user already identified and tracked in the camera frame.

Table 23: REQ-ST-04

TEST CASES	INPUT	INITIAL STATE(S)	EXPECTED OUTPUT STATE	EXPECTED VARIABLE CHANGE(S)	ACTUAL OUTPUT STATE	ACTUAL VARIABLE CHANGE(S)	PASS
ST-04-T01	Gesture 2	Idle	Follow	None	Follow	None	Yes
ST-04-T02	Gestures 1,3,4,5	Idle	Idle	None	Idle	None	Yes

Pass/Fail Criteria

A test is considered a Pass if the system transitions to the Follow state only when it is currently in the Idle state and the tracked user provides the valid follow gesture. All other gestures must be ignored for follow-state transition purposes, and the system must remain in Idle. Any unintended transition to Follow is considered a Fail.

REQ-ST-05: The system shall not transition from the follow state to the idle state unless the tracked user is lost or the tracked user provides the stop follow gesture.

Test description: Provide various commands and have the user leave the frame while tracked and in the follow state to evaluate the requirement

Test setup: System shall first be brought into the Follow state with a valid tracked target user in frame. The tester shall then provide different inputs in separate trials, including the stop-follow gesture, non-stop gestures, and leaving the camera frame to simulate target lost. For the target lost case, the tracked user shall move completely out of the camera field of view or be occluded long enough for tracking to be lost.

Table 24: REQ-ST-05

TEST CASES	INPUT	INITIAL STATE(S)	EXPECTED OUTPUT STATE	EXPECTED VARIABLE CHANGE(S)	ACTUAL OUTPUT STATE	ACTUAL VARIABLE CHANGE(S)	PASS
ST-05-T01	Gesture 1	Follow	Idle	C_linear_velocity_x: -> 0 C_angular_velocity_z: -> 0	Idle	C_linear_velocity_x: 0.2 -> 0	Yes
ST-05-T02	Gesture 2	Follow	Follow	None	Follow	None	Yes
ST-05-T03	Gesture 3	Follow	Follow	C_distance += 0.25m			Incomplete
ST-05-T04	Target Lost	Follow	Idle	C_linear_velocity_x: -> 0	Idle	C_linear_velocity_x: 0.2 -> 0	Yes

Pass/Fail Criteria

A test is considered a Pass if the system transitions from the Follow state to the Idle state only when a valid stop gesture is provided by the tracked user or when the tracked user is lost. All other gestures must not trigger a transition. The system must stop motion when transitioning to Idle. Any unintended transition or continued motion is considered a Fail.

REQ-ST-06: The system shall transition to manual mode from any state given the user provides the manual mode input signal.

& REQ-ST-07: The system shall not transition to manual mode unless the user provides the manual mode input signal.

Test description: Enter the manual mode toggle command while in various states to evaluate the requirement.

Test setup: System shall be placed in different starting states as required by the test cases, including Initial, Idle, and Follow. The tester shall apply the manual mode toggle command while the system is in each state. For comparison, no-toggle conditions may also be observed to verify that no transition to Manual Mode occurs without the manual mode input signal. MATLAB or the UI software shall be used to confirm state transitions and monitor the value of C_mm_toggle.

Table 25: REQ-ST-06

TEST CASES	INPUT	INITIAL STATE(S)	EXPECTED OUTPUT STATE	EXPECTED VARIABLE CHANGE(S)	ACTUAL OUTPUT STATE	ACTUAL VARIABLE CHANGE(S)	PASS
ST-06/07-T01	Manual mode toggle off to on	Initial	Manual	C_mm_toggle 0 -> 1	Manual	C_mm_toggle 0 -> 1	Yes
ST-06/07-T02	Manual mode toggle off to on	Idle	Manual	C_mm_toggle 0 -> 1	Manual	C_mm_toggle 0 -> 1	Yes
ST-06/07-T03	Manual mode toggle off to on	Following	Manual	C_mm_toggle0 -> 1	Manual	None	Yes

Pass/Fail Criteria

A test is considered a Pass if the system transitions to Manual Mode from any state only when the manual mode toggle signal is provided. In the absence of the toggle signal, the system must remain in its current state. Any unintended transition to Manual Mode is considered a Fail.

REQ-ST-08: The system shall not exit manual mode unless the user provides the manual mode exit signal.

& REQ-ST-09: Provided the exit condition is met, the system shall transition from manual mode to the initial state if there is no current tracked person.

Test description: Enter the manual mode toggle command while in the manual mode state to evaluate the requirement.

Test setup: System state shall be set to Manual Mode. The tester shall apply the manual mode exit signal by toggling manual mode off.

Table 26: REQ-ST-08&09

TEST CASES	INPUT	INITIAL STATE(S)	EXPECTED OUTPUT STATE	EXPECTED VARIABLE CHANGE(S)	ACTUAL OUTPUT STATE	ACTUAL VARIABLE CHANGE(S)	PASS
ST-08/09-T01	Manual mode toggle on to off and M_p_class = 0	Manual	Initial	C_mm_toggle 1 -> 0	Initial	C_mm_toggle 1 -> 0	Yes
ST-08/09-T02	Manual mode toggle on to off and M_p_class = 1	Manual	Idle	C_mm_toggle 1 -> 0	Idle	C_mm_toggle 1 -> 0	Yes

Pass/Fail Criteria

A test is considered a Pass if the system exits Manual Mode only when the manual mode exit signal is provided. After exit, the system shall transition to Initial when no tracked person is present, and to Idle when a tracked person is present. In the absence of the exit signal, the system must remain in Manual Mode. Any unintended exit or incorrect next state is considered a Fail.

REQ-ST-10: Provided the exit condition is met, the system shall transition from manual mode to the idle state if there is a current tracked person.

Test description: Toggle manual mode off while a target person is tracked to evaluate the requirement.

Test setup: System state shall be set to Manual Mode. A tracked target user shall be present in the camera frame for the case where M_p_class = 1, and absent from the frame for the case where M_p_class = 0. The tester shall then toggle manual mode off to trigger the exit condition. MATLAB or the UI software shall be used to monitor person-tracking status, C_mm_toggle, and the resulting transition to Idle or Initial as appropriate.

Table 27: REQ-ST-10

TEST CASES	INPUT	INITIAL STATE(S)	EXPECTED OUTPUT STATE	EXPECTED VARIABLE CHANGE(S)	ACTUAL OUTPUT STATE	ACTUAL VARIABLE CHANGE(S)	PASS
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ST-10-T01	(c_mm_toggle == 0) & (m_p_class == 1)	Manual Mode	Idle	C_mm_toggle 1 -> 0	Idle	C_mm_toggle 1 -> 0	Yes
ST-10-T02	(c_mm_toggle == 0) & (m_p_class == 0)	Manual Mode	Initial	C_mm_toggle 1 -> 0	Initial	C_mm_toggle 1 -> 0	Yes

Pass/Fail Criteria

A test is considered a Pass if, upon exiting Manual Mode, the system transitions to the Idle state when a tracked person is present, and to the Initial state when no tracked person is present. The exit condition must be triggered by the manual mode toggle signal. Any incorrect resulting state is considered a Fail. This test verifies correct state recovery after manual override based on real-time perception of user presence.

5.2.3 General State Behaviour

REQ-GSB-01: The system shall provide a **Manual Mode** where the user can directly control the PT through DT
& REQ-GSB-03: The system shall enable the user to switch between "Autonomous" and "Manual" modes at any time without restarting the system.

Test description: Refer to test cases below

Table 28: REQ-GSB-01 & REQ-GSB-03

TEST CASES

SB-06-T01

SB-06-T02

SB-06-T03

SB-06-T04

SB-06-T05

SB-06-T06

ST-06/07-T01

ST-06/07-T02

ST-06/07-T03

ST-08/09-T01

ST-08/09-T02

ST-10-T01

ST-10-T02

REQ-GSB-02: The system shall allow the user to interrupt autonomous operation at any time via a hardware or software command, transitioning the vehicle to a **STOP** (initial) state within **200ms**.

Test description: Provide the emergency stop command in all states to evaluate the requirement.

Test setup: System shall be placed in each relevant operating state required by the test cases, including Manual, Follow, and Idle. While the system is operating, the tester shall activate the emergency stop command through the available hardware or software emergency-stop mechanism. If the PT is moving, it shall be commanded to move before the emergency stop is applied so that stopping behaviour can be observed. MATLAB, ROS2 timestamps, or the UI software shall be used to monitor state transition to Initial, command outputs, and whether linear/angular velocities are reduced to zero within the required response time.

Table 29: REQ-GSB-02

TEST CASES	INPUT	INITIAL STATE(S)	EXPECTED OUTPUT STATE	EXPECTED VARIABLE CHANGE(S)	ACTUAL OUTPUT STATE	ACTUAL VARIABLE CHANGE(S)	PASS
GSB-02-T01	C_emergency_stop	Manual	Initial	C_linear_velocity_x: -> 0	Initial	C_linear_velocity_x: -> 0	Yes
				C_angular_velocity_z: -> 0		C_angular_velocity_z: -> 0	
GSB-02-T02	C_emergency_stop	Following	Initial	C_linear_velocity_x: -> 0	Initial	C_linear_velocity_x: -> 0	Yes
				C_angular_velocity_z:			

				-> 0		C_angular_velocity_z:	
						-> 0	
GSB-02-T03	C_emergency_stop	Idle	Initial	None	Initial	None	Yes

5.2.4 System Monitoring

The system monitoring requirements 1 & 3 are based on visual inspections of the user interface, which will be a different testing method than other requirements. We plan to let the system operate in one simple cycle(Follow the target user and read all possible gestures) in different environments with the default variables and inspect the changes of the monitored variables and their delay time.

REQ-SM-01: The DT shall provide a visual interface to monitor the real-time state variables.

& REQ-SM-02: The DT shall mirror the states and the monitored variables of the PT **with a maximum average latency of 500ms**

Test description: Operate the system in a normal way for both manual mode and following mode, observe the visual interface to evaluate the requirement.

Table 30: REQ-SM-01&02

TEST CASES	INPUT	STATE CHANGES	VELOCITY GRAPHS	VIDEO UPDATES	YOLO BOUNDING BOX	LIDAR MAP	PASS
SM-01/02-T01	Simple operation cycle 1 (power on, track target	Yes	Yes	Yes	Yes	No	Incomplete

	user, target user gives follow gesture and moves around)							
SM-01/02-T02	Simple operation cycle 2 (power on, enter manual mode and move around)	Yes	Yes	Yes	Yes	No	Incomplete	

6 Appendix

6.1 Referenced Figures

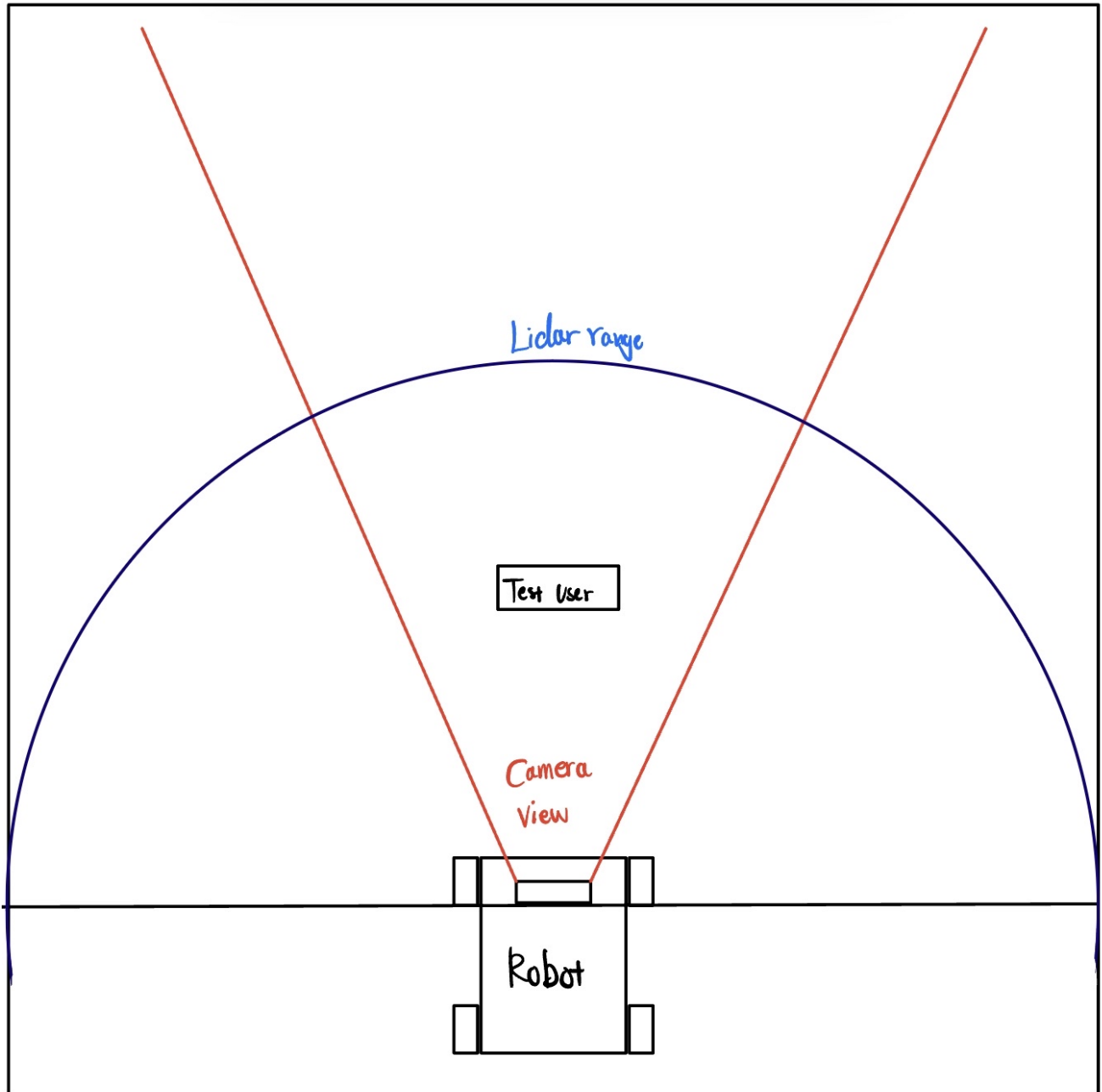


Figure 1: REQ-SB-01 test visual description.

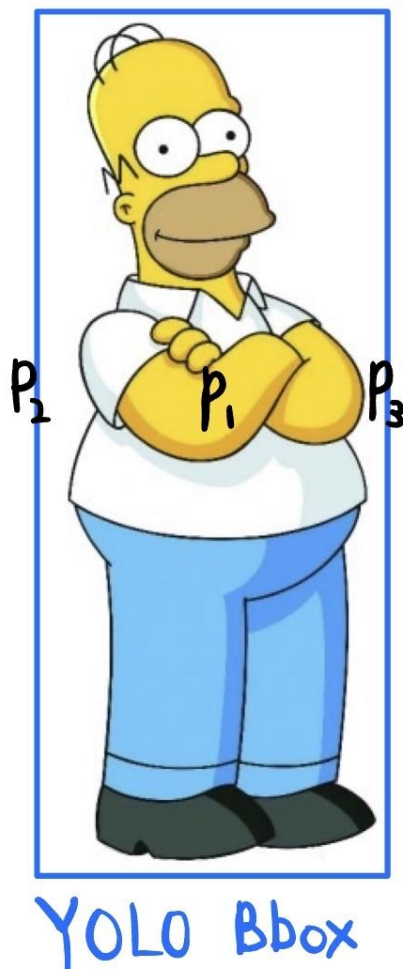


Figure 2: Visual representation of YOLO Box and Gesture Location

6.2 Previous Requirements from System Requirements & Hazard Analysis

6.2.1 PT-X

PT-1. The core motion-control loop (reading monitored variables, computing commands, and updating controlled variables) shall run with a period of no more than 0.1 s (i.e., at least 10 Hz) during Follow mode. This ensures that changes in the target person's motion and nearby obstacles are reflected in vehicle commands quickly enough to feel smooth and safe.

PT-2. Newly observed obstacle information (for example, from M_Scan, M_Point, or M_Depth) shall be considered in motion decisions within 0.1 s of being acquired. This aligns with DSR2 so that the vehicle reacts to new obstacles before entering unsafe regions.

PT-3. Under nominal conditions, the end-to-end latency from a valid change in gesture (M_YOLOGesture) to the corresponding mode change (C_Mode) taking effect shall not exceed 0.5 s. This keeps the system responsive to user commands and avoids confusing delays between a gesture and the observed change in behavior.

PT-4. When the target position (M_TargetPosition or M_DistanceApart) changes significantly, the resulting changes in C_Velocity and C_Steering shall occur with an end-to-end latency of no more than 0.3 s. This ensures that following behavior remains natural and prevents overshoot or oscillatory motion due to stale information.

PT-5. If the system becomes uncertain about the target identity or position, it shall detect this condition within 1.0 s and then decelerate to a complete stop within an additional 0.5 s, entering a “Target Lost” state, as required by DSR8. This limits the distance the vehicle can travel while tracking is unreliable and reduces the risk of following the wrong person.

PT-6. If any required sensor (such as LiDAR or camera) fails to update for longer than its specified timeout, the system shall transition to a safe STOP state within 0.2 s of detecting the timeout. This prevents the controller from making motion decisions based on missing or outdated sensor data.

PT-7. During Follow mode, the system shall log in key monitored and controlled variables at a rate of at least 5 Hz. This supports post-hoc analysis and verification of safety requirements and timing behavior.

PT-8. The emergency stop channel (hardware button) shall cut power to the drive actuators within 200 ms of activation,

6.2.3 HW-X

HW-1. The physical twin shall include at least two independent sensing modalities for obstacle detection (for example, a range-based sensor and a vision-based method), providing coverage within a radius of at least 2 m around the vehicle. This supports DSR1 by ensuring redundancy in obstacle perception.

HW-2. The obstacle-detection sensors shall provide range measurements with a resolution of no worse than 0.05 m and an update rate of at least 5 Hz in the primary operating region in front of the vehicle. This enables reliable detection of nearby obstacles in time to react safely.

HW-3. The system shall include at least one RGB or RGB-D camera whose field of view is sufficient to fully contain the expected target person at typical following distances (0.5–2 m), with a nominal frame rate of at least 10 fps under indoor lighting. This provides adequate visual coverage and temporal resolution for person and gesture perception.

HW-4. The drive system shall support independent control of forward velocity and steering (or equivalently, differential wheel speeds), with command resolution fine enough that each command step can change velocity by no more than 0.05 m/s and steering angle by no more than 0.05 rad. This allows the controller to implement smooth, incremental adjustments to heading and speed.

HW-5. The vehicle shall be equipped with wheel encoders and/or an IMU capable of estimating linear velocity with an error of no more than 0.05 m/s over one-second intervals. This supports closed-loop feedback and detection of actuator faults in line with DSR3.

HW-6. A hardware emergency stop mechanism shall be physically accessible to the operator while walking near the vehicle and shall interrupt power to the drive actuators independently of the main control computer. This provides a last-resort safety mechanism that does not rely on software.

HW-7. The on-board power system (battery and power distribution) shall supply stable voltage and current sufficient for continuous operation of all sensors, computer hardware, and actuators for at least the planned demonstration duration. This reduces the risk of unexpected brownouts or sudden shutdowns that could lead to unsafe behavior.

HW-8. The communication link between the physical and digital twin (for example Wi-Fi or an equivalent technology) shall support a sustained data rate sufficient to carry all required sensor streams and control commands while respecting the timing requirements defined in Section 7.1. This ensures that remote computation does not violate latency and update-rate constraints.

HW-9. All safety-critical connections, such as the emergency stop circuitry, motor driver enable lines, and battery monitoring connections, shall be electrically robust and shall be designed to tolerate single-point faults such as loose connectors without silently disabling safety functions. This reduces the risk that a simple hardware fault prevents safe operation.

6.2.2 SW-X

SW-1. The runtime environment shall provide a publish–subscribe communication middleware (for example ROS 2) that supports reliable message passing for all monitored and controlled variables between the physical twin and the digital twin. This enables perception, control, and GUI components to exchange data in a modular and observable way.

SW-2. The operating system shall provide a system clock with a resolution of at most 1 ms, and the software shall timestamp sensor messages and control commands using this clock. This allows the system to measure and enforce the timing requirements described in Section 7.1.

SW-3. The software environment shall support deep-learning inference for person and gesture detection on the incoming camera stream at a sustained effective rate of at least 5 Hz under nominal conditions. This ensures that M_YOLOPerson and M_YOLOGesture are updated frequently enough for responsive following and mode switching.

SW-4. The middleware shall support quality-of-service (QoS) configuration or equivalent mechanisms that allow safety-critical topics (such as obstacle data, emergency stop, and velocity commands) to be prioritized over non-critical topics (such as the GUI video feed) when computing or communication resources are limited. This ensures that degraded performance affects non-critical functions first, thereby preserving safety.

SW-5. The software stack shall provide a configurable logging and replay facility capable of recording key monitored and controlled variables at the rates specified in PT-8. This enables debugging, verification of safety behavior, and offline analysis of incidents.

SW-6. The GUI (dashboard) shall run as a separate process or node that subscribes to system state topics without blocking the real-time control and perception loops. This prevents visualization tasks from interfering with safety-critical timing.

SW-7. The environment shall provide a simulation or test harness, such as playback of logged data or a simple virtual environment, that allows the control logic and safety requirements to be exercised without requiring a physical vehicle. This facilitates early validation of requirements and regression testing as the design evolves.

6.2.4 Derived Safety Requirements (DSR)

DSR1: The vehicle shall use at least two independent sensing modalities for obstacle detection. The loss of any single sensing channel does not prevent it from detecting obstacles correctly.

DSR2: The vehicle shall take newly observed obstacles into account in its motion decisions within 0.1 s. Otherwise, it will reduce speed or even come to an immediate stop.

DSR3: If the vehicle's actual speed deviates significantly from its intended motion for more than 0.5 s, it shall come to a controlled stop and further motion shall be stopped until the fault is cleared.

DSR4: The "emergency stop" (hardware button) shall cut power to the drive actuators within 200 ms, regardless of current operating mode or software state.

DSR5: The vehicle shall not enter a region which has not been verified confidently. If no such safe path exists, the vehicle shall remain stationary.

DSR6: The vehicle shall automatically reduce speed when the motion control updates cannot be maintained at a required frequency.

1.2 Loss of Tracking

Risk Mitigation Strategies:

- Provide multi-modal target identification.
- Implement a "target lost" handling state instead of continuing the motion with uncertainty.
- Battery-aware follow behavior
- Constrain camera mounting and FOV to handle "camera positioned incorrectly" or "blocked by moving objects"

Derived Safety Requirements:

DSR7: The vehicle should identify the target person with a reliability sufficient to prevent following a wrong person. If the system cannot confidently identify the target person, it will not initiate.

DSR8: If the vehicle becomes uncertain about the target's identity or position for more than 1.0 s, it shall decelerate and come to a complete stop within 0.5 s, entering a "Target Lost" state.

DSR9: If the target is not visible in the camera's intended FOV for longer than 1.5 s, the vehicle will not move toward the last predicted position until the target is redefined.

DSR10: The vehicle shall terminate the follow mode with a controlled stop and user notification when battery capacity becomes critically low.

DSR11: If the target's movement cannot be predicted reliably, the vehicle shall stop and remain stationary until reliable information is restored.

1.3 Battery Fire

Risk Mitigation Strategies:

- Contain battery-related hazards (thermal, mechanical, electrical) using physical protection.
- Detect communication loss and default to a controlled safe state rather than continuing operation.

Derived Safety Requirements:

DSR12: If the vehicle detects any abnormal battery condition, it shall immediately stop motion, disable charging/discharging operations, and switch to a safe shutdown state with a user warning.

DSR13: If the vehicle loses required communication with the host computer or essential control devices, it shall stop all motion and enter a pending state until communication is restored.

DSR14: The vehicle shall disable operation and require manual recovery whenever a fault prevents safe continued operation.